

Sparse cyclic Weyl operators in prime-power dimension: an operator extension of the scalar uncertainty law through sparsity $2p - 1$

Lluis Eriksson (Independent researcher)

14 September 2026 — revised candidate

Abstract

Let $d = p^k$ and let $W_{(a,b)} = X^a Z^b$ be the cyclic Weyl operators on $\mathbb{C}^d$. Write $w_p(r)$ for the base- p digit weight $\prod_t (r_t + 1)$ and $\Omega_k(m) = \max\{r < p^k : w_p(r) \leq m\}$. We give a valuation proof of the classical Delvaux–Van Barel scalar law: (Z) a polynomial with at most m terms and degree below d annihilates at most $\Omega_k(m)$ of the d -th roots of unity, and this is sharp. Our operator extension is: (B) every nonzero combination C of $\ell \leq 2p - 1$ Weyl operators satisfies $\dim \ker C \leq \Omega_k(\ell)$, i.e. $\text{rank } C \geq p^k - (\ell - 1)p^{k-1}$ for $\ell \leq p$ (sharp) and $\text{rank } C \geq p^{k-1}$ for $p \leq \ell \leq 2p - 1$. The commutative ingredient is the Massey–Costello–Justesen weight-retaining property; the operator ingredient is a Radon-type lemma showing that first-order obstructions have weight at least $2p$. For composite d the maximal nullity is $d/p_{\min}$ at sparsity two and, conditionally on a published trinomial rank rule, $\max(2d/r_3, d/2 [d \text{ even}])$ at sparsity three. The conjecture $\dim \ker C \leq \Omega_k(s_W(C))$ for all sparsities is proved for $d \in \{4, 8, 9\}$ and for torsion cosets; its first open case is $d = 16, \ell = 6$.

1. Introduction

1.1 The results

Fix $d = p^k$, $\omega = e^{2\pi i/d}$, the cyclic Weyl basis $W_{(a,b)} = X^a Z^b$ of $M_d(\mathbb{C})$ indexed by $G = (\mathbb{Z}/d)^2$, and for $C = \sum_g c_g W_g$ the Weyl sparsity $s_W(C) = \#\{g : c_g \neq 0\}$. The published prime law [3W] states that in prime dimension $\dim \ker C \leq s_W(C) - 1$. Its Open question 3 asked what replaces this in prime-power dimension, where the coranks $p^j, 2p^j$ of [0WE] Theorem 3 already show that $\ell - 1$ is wrong, and suggested the candidate $(\ell - 1)p^{k-1}$. This paper answers the question in the range $\ell \leq 2p - 1$ and identifies the function that governs the problem. With $w_p(r) = \prod_t (r_t + 1)$ for $r = \sum_t r_t p^t$ and

$$\Omega_k(m) = \max\{r < p^k : w_p(r) \leq m\},$$

the scalar antecedent and the operator results are:

- **Theorem Z (Delvaux–Van Barel scalar law, in digit-weight notation).** The maximal number of p^k -th roots of unity annihilated by a nonzero polynomial of degree $< p^k$ with at most m nonzero coefficients is exactly $\Omega_k(m)$. The product construction used in our proof is certified exactly in $\mathbb{Z}[\omega_d]$ for $d \in \{4, 8, 9, 16, 25, 27, 32, 49, 81\}$.
- **Theorem B (operators).** If $d = p^k$ and $1 \leq s_W(C) = \ell \leq 2p - 1$, then $\dim \ker C \leq \Omega_k(\ell)$. Explicitly, $\dim \ker C \leq (\ell - 1)p^{k-1}$ for $\ell \leq p$, sharp at every such ℓ , and $\dim \ker C \leq (p - 1)p^{k-1}$

for $p \leq \ell \leq 2p - 1$ ($k \geq 2$). The candidate $(\ell - 1)p^{k-1}$ of [3W] is therefore right exactly up to $\ell = p$; beyond it the true envelope is Ω_k , which grows much more slowly (for example $\Omega_k(p^a) = p^k - p^{k-a}$).

- **Theorems T2 and T3 (composite d).** For every $d \geq 2$ the maximal nullity at sparsity two is $d/p_{\min}(d)$; for every $d \geq 3$ the maximal nullity at sparsity three is $\max(2d/r_3(d), \frac{d}{2}[d \text{ even}])$, where $r_3(d)$ is the smallest divisor of d that is at least 3; the second branch is attained by an anticommuting pair when $d \equiv 2 \pmod{4}$. Theorem T3 is conditional on the trinomial rank rule [0WE] Theorem 2, used as published. Both laws refute every additive law in the largest prime or prime-power factor of d .

The commutative ingredient (Theorem A, Section 3) is a characteristic- p statement: a polynomial with m nonzero terms vanishes at $x = 1$ to order r only if $w_p(r) \leq m$. This is a special case of the weight-retaining property of Massey, Costello and Justesen [MCJ73], and we say so plainly; a short self-contained proof is included because the form needed here (arbitrary fields of characteristic p) is the one used in the valuation argument. The operator ingredient is new relative to [3W]: the pigeonhole over $p + 1$ directions of [3W] is replaced by Lemma N, which shows that a residue vector with vanishing fibre sums in all $p^{k-1}(p + 1)$ primitive directions has at least $2p$ nonzero entries, via a Radon-type lemma on $\mathbb{F}_p^2$ (Lemma R).

Beyond $2p - 1$ the first-order method genuinely stops (two parallel lines of a p -torsion coset form an obstruction of weight $2p$), and we state Conjecture C: $\dim \ker C \leq \Omega_k(s_W(C))$ for all C . It is a theorem for $d \in \{4, 8, 9\}$, for every support inside a coset of the p -torsion, for every support inside a coset of the 4-torsion when $d \in \{16, 32\}$, and wherever the field-free bound $\text{rank } C \cdot s_W(C) \geq d$ suffices; the first open case is $d = 16, \ell \in \{6, 7\}$ (Section 6).

1.2 Antecedents

The prime case is [3W] (Sections 2–5 there: the setting, the valuation reduction modulo $(1 - \omega)$, the explicit Clifford conjugations $F'Q_s$, the pigeonhole, and Frenkel’s lemma), which in turn rests on Tao’s uncertainty principle [Ta05] and Frenkel’s proof of Chebotarev’s theorem [Fr04]. The three-term record [0WE] proves the trinomial rank rule (Theorem 2 there, with its Lemma 1 on central sectors), the classification of coranks for $s_W \leq 3$ in prime-power dimension (Theorem 3), and exhibits composite counterexamples to any $\ell - 1$ law: $2^{1/4}I + X^3 + Z^3$ in $d = 12$ (nullity 3, Proposition 4), $I + X^3 + iX^{-3}Z^{18}$ in $d = 36$ (nullity 6, eq. (6a)), and the binomial $X(I + Z^6)$ in $d = 12$ (nullity 6, eq. (6c)). The origin of the Weyl rank problem is quantum list decoding: [3H0] determines exact list thresholds for Weyl channels with consecutive and arithmetic probes, [0WE] eq. (8) reduces list decoding to a rank problem, and [3W] Corollary 2 closes the prime case. The list-decoding consequences of the present rank bounds are not pursued here.

On the commutative side, Meshulam’s uncertainty inequality for finite abelian groups [Me06] is the general tool (used in Section 6 for torsion cosets); Delvaux and Van Barel [DVB08] prove the prime-power Fourier Hamming-number law, which is exactly the monotone scalar zero-count envelope of Theorem Z in equivalent notation (Definition 7 and Theorem 10, equation (23), in their author-hosted report [DVB06]); Lam and Leung [LL00] classify the lengths of vanishing sums of roots of unity, the phenomenon behind the “smallest divisor” laws of Section 5; B  r   and Lev [BL18] give the sharp uncertainty inequality on $\mathbb{F}_p^2$, relevant to the p -torsion cosets of Section 6; Loukaki [Lo25] treats principal Fourier minors of composite order pq . Lucas’ theorem [Lu78] is what makes w_p the right weight.

1.3 Calibration

Theorem A is classical [MCJ73], and the scalar envelope Theorem Z is the known Delvaux–Van Barel theorem [DVB06, DVB08]; Section 4.2 supplies a self-contained proof through valuation reduction and digit weight. Lemma R is the classical minimum-distance statement for the dual incidence code of the affine plane of prime order, stated here over any characteristic- p field with an elementary proof; see [KMM08, Section 2, Result 1]. The contribution proposed here is the extension to arbitrary complex combinations of cyclic Weyl operators through sparsity $2p - 1$, the valuation formulation B', and the primitive-direction/torsion-coset mechanism N. The search for prior operator results is bounded and does not certify worldwide priority. Theorems T2 and T3 are short consequences of trace orthogonality and of the imported [OWE] rank rule, respectively; T3 remains conditional. Verification coverage is itemised in Section 7. The supplied history includes author-only runs, separately implemented checks and unrepeated computations, not a universal independent replay.

2. Notation and setting

Weyl operators. For $d \geq 2$ let $\omega = \omega_d = e^{2\pi i/d}$, $X|j\rangle = |j+1 \bmod d\rangle$, $Z|j\rangle = \omega^j|j\rangle$, so $ZX = \omega XZ$, and $W_{(a,b)} = X^a Z^b$ for $(a,b) \in G = (\mathbb{Z}/d)^2$. Then $W_g W_h = \omega^{\langle g,h \rangle} W_{g+h}$ with $\langle (a,b), (a',b') \rangle = ba'$, $\text{Tr } W_g = 0$ for $g \neq 0$, and the W_g form a trace-orthogonal basis of $M_d(\mathbb{C})$. For $C = \sum_g c_g W_g$ the support is $S(C) = \{g : c_g \neq 0\}$ and $s_W(C) = |S(C)|$. Left multiplication by W_h^{-1} translates the support by $-h$, multiplies coefficients by roots of unity and preserves the kernel; conjugation by an invertible matrix U with $UW_g U^{-1} \in \mathbb{C} W_{M_g}$ ($M \in SL_2(\mathbb{Z}/d)$) maps the support to MS and preserves rank. Two families of such U are made explicit in Section 4.3.

Digit weight. For $d = p^k$ and $0 \leq r < p^k$ write $r = \sum_{t=0}^{k-1} r_t p^t$ with $0 \leq r_t < p$ and put $w_p(r) = \prod_t (r_t + 1)$. By Lucas' theorem, $\binom{r}{i} \not\equiv 0 \pmod{p}$ exactly when every digit of i is at most the corresponding digit of r , so $w_p(r)$ is the number of nonzero terms of $(x-1)^r$ in characteristic p . For $m \geq 1$ let $\Omega_k(m) = \max\{r < p^k : w_p(r) \leq m\}$; it is nondecreasing in m , $\Omega_k(1) = 0$, $\Omega_k(p^k) = p^k - 1$, and

$$\begin{aligned} \Omega_k(m) &= (m-1)p^{k-1} & (1 \leq m \leq p), \\ \Omega_k(m) &= (p-1)p^{k-1} + ([m/p] - 1)p^{k-2} & (p \leq m < p^2, k \geq 2), \\ \Omega_k(p^a) &= p^k - p^{k-a} & (0 \leq a \leq k). \end{aligned}$$

(For $m \leq p$: a number above $(m-1)p^{k-1}$ has top digit $\geq m$, or top digit $m-1$ and a further nonzero digit, hence weight $> m$. The other two formulas are checked the same way; all three were also verified by computer for every $p^k \leq 2401$.) For example, for $d = 16$ the values $\Omega_4(1..16)$ are 0, 8, 8, 12, 12, 12, 12, 14, ..., 14, 15; for $d = 27$, $\Omega_3(m)$ is 0, 9, 18 for $m = 1, 2, 3$, then 18 ($m = 4, 5$), 21 ($6 \leq m \leq 8$), 24 ($9 \leq m \leq 17$), 25 ($18 \leq m \leq 26$), 26.

Torsion and directions. For $d = p^k$ let $T_j = p^{k-j}G$ be the p^j -torsion, so $T_1 \cong \mathbb{F}_p^2$. A *primitive direction* is a surjective linear form $\lambda : G \rightarrow \mathbb{Z}/d$ up to units; there are $|\mathbb{P}^1(\mathbb{Z}/p^k)| = p^{k-1}(p+1)$ of them, and the fibres of λ are the cosets of $\ker \lambda$, a cyclic subgroup of order p^k ; we call these cosets *primitive lines*. Every cyclic subgroup of order p^k is the kernel of exactly one primitive direction. A support contained in a primitive line is called *collinear*.

Valuation rings. Let K be a subfield of $\mathbb{C}$ containing ω_d . A valuation ring $\mathcal{O} \subset K$ with maximal ideal $\mathfrak{m}$, residue field $\mathbb{k} = \mathcal{O}/\mathfrak{m}$ and valuation v is *admissible* if $1 - \omega_d \in \mathfrak{m}$. Since $1 - \omega_{p^k}$ generates the unique prime of $\mathbb{Z}[\omega_{p^k}]$ above p and p is a unit multiple of $(1 - \omega_{p^k})^{\varphi(p^k)}$, admissibility forces $p \in \mathfrak{m}$, so $\text{char } \mathbb{k} = p$ and every d -th root of unity reduces to 1. Admissible valuation rings exist on every finitely

generated $K \supseteq \mathbb{Q}(\omega_d)$ by Chevalley's extension theorem ([St, Tag 00IA]; this is the route of [3W], Step 1). Two rank facts are used: for $A \in M_d(\mathcal{O})$, $\text{rank}_K A \geq \text{rank}_{\mathbb{K}} \bar{A}$; for $A \in M_d(K)$, $\text{rank}_K A = \text{rank}_{\mathbb{C}} A$.

Root-of-unity counts. For any d and $m \geq 1$ put $Z(d, m) = \max \#\{\zeta \in \mu_d : f(\zeta) = 0\}$ over polynomials f with exactly m nonzero coefficients and exponents distinct modulo d , and $Z_{\leq}(d, m) = \max_{m' \leq m} Z(d, m')$. Since $\{f(\zeta)\}_{\zeta \in \mu_d}$ is the spectrum of $f(Z)$, $Z(d, m)$ is the maximal nullity of a polynomial in Z with exactly m terms, i.e. of a collinear Weyl combination on the Z -axis with exact sparsity m .

3. Statements

Theorem A (order and digit weight in characteristic p ; [MCJ73]). Let $\mathbb{k}$ be any field of characteristic p and $f \in \mathbb{k}[x]$ nonzero with exactly m nonzero monomials. Then $w_p(\text{ord}_{x=1} f) \leq m$. If moreover $\deg f < p^k$, then $\text{ord}_1 f \leq \Omega_k(m)$, and $\max\{\text{ord}_1 f : \deg f < p^k, \# \text{terms} \leq m\} = \Omega_k(m)$, attained by $(x-1)^{\Omega_k(m)}$.

Theorem A is the weight-retaining property of Massey, Costello and Justesen ([MCJ73], Theorem 1.1 for $\text{GF}(2^r)$ and Section III-B, Theorem 6.1 for $\text{GF}(q)$, with their Lemma 1 giving $W[(x-c)^i] = \prod_j (i_j + 1)$): writing $f = \sum_{i \geq r} b_i (x-1)^i$ with $b_r \neq 0$ and $r = \text{ord}_1 f$, their theorem states that the Hamming weight of f is at least that of $(x-1)^r$. They state it over finite fields; the proof in Section 4.1 is written for an arbitrary field of characteristic p , which is the form the valuation argument needs. The degree restriction is needed only for the reformulation through Ω_k (note $x^{p^k} - 1 = (x-1)^{p^k}$ has two terms).

Theorem Z (Delvaux-Van Barel scalar law in equivalent notation). For $d = p^k$ and $m \geq 1$, $Z_{\leq}(d, m) = \Omega_k(m)$. Equivalently, a polynomial in Z with at most m terms has nullity at most $\Omega_k(m)$, and for every $r < p^k$ the polynomial

$$f_r(z) = \prod_{j=1}^k \prod_{i=1}^{r_{j-1}} (z^{p^{j-1}} - \omega_p^i), \quad \omega_p = \omega^{p^{k-1}} = e^{2\pi i/p}, \quad r = \sum_{j=1}^k r_{j-1} p^{j-1},$$

has exactly $w_p(r)$ nonzero coefficients, degree r , and exactly r zeros in μ_d . In particular $Z(d, m) = \Omega_k(m)$ whenever $w_p(\Omega_k(m)) = m$ (for instance $m \leq p$), and $Z(d, m) \leq \Omega_k(m)$ always.

Equivalence with the classical Hamming-number formula. Let $1 \leq m < p^k$, and write $cp^a \leq m < (c+1)p^a$, with $1 \leq c \leq p-1$ and $0 \leq a < k$. Then

$$\Omega_k(m) = p^k - (p-c+1)p^{k-a-1}, \quad H_{F_{p^k}}(m) = p^k - \Omega_k(m).$$

The first integer has a leading base- p digits $p-1$, then $c-1$, then zeros, so its weight is cp^a . Any larger integer has digit weight at least $(c+1)p^a$: either an earlier digit increases, or a lower nonzero digit at least doubles the weight, and $2c \geq c+1$. The second identity is [DVB06, Definition 7 and Theorem 10, equation (23)]: their support budget is at most m . Fourier values are polynomial evaluations on roots of unity, so minimum nonzero Fourier support equals p^k minus maximum zero count. For $m \geq p^k$, separately, $\Omega_k(m) = p^k - 1$ and the minimum Fourier support is 1.

Theorem B' (valuation form, all sparsities). Let $d = p^k$, let $K \subseteq \mathbb{C}$ contain ω_d and, if $p = 2$, also ω_{2d} ; let $\mathcal{O} \subset K$ be admissible. Let $C = \sum_{g \in S} c_g W_g$ with all $c_g \in K^\times$, scaled so that $\min_g v(c_g) = 0$, and let $\bar{c}_g \in \mathbb{k}$ be the residues. For a primitive direction λ put $\gamma_t^\lambda = \sum_{\lambda(g)=t} \bar{c}_g$ ($t \in \mathbb{Z}/d$) and $m_\lambda = \#\{t : \gamma_t^\lambda \neq 0\}$. Then for every λ with $m_\lambda \geq 1$,

$$\dim \ker C \leq \Omega_k(m_\lambda) \leq \Omega_k(|S|).$$

Lemma R (Radon kernel on $\mathbb{F}_p^2$). Let $\mathbb{k}$ have characteristic p and let $f : \mathbb{F}_p^2 \rightarrow \mathbb{k}$ be nonzero with zero sum on every affine line. Then $|\text{supp } f| \geq 2p$. The bound is attained by $1_{L_1} - 1_{L_2}$ for two parallel lines.

Lemma N (first-order obstructions have weight at least $2p$). Let $d = p^k$ and let $\bar{c} : G \rightarrow \mathbb{k}$ be nonzero with zero sum on every primitive line. Then $|\text{supp } \bar{c}| \geq 2p$; two parallel lines inside a coset of T_1 attain the bound.

Theorem B (operator law in prime-power dimension, $\ell \leq 2p - 1$). Let $d = p^k$ and $C \neq 0$ with $s_W(C) = \ell \leq 2p - 1$. Then $\dim \ker C \leq \Omega_k(\ell)$; explicitly

$$\text{rank } C \geq p^k - (\ell - 1)p^{k-1} \quad (\ell \leq p), \quad \text{rank } C \geq p^{k-1} \quad (p \leq \ell \leq 2p - 1, k \geq 2).$$

For $\ell \leq p$ the bound is attained by $C_\ell = \prod_{i=0}^{\ell-2} (Z^{p^{k-1}} - \omega_p^i I)$, which has $s_W = \ell$ and nullity $(\ell - 1)p^{k-1}$. For $k = 1$ this is [3W] Theorem 1.

Corollary (coranks at sparsity ≤ 3). $\{r : w_p(r) \leq 3\} = \{0\} \cup \{p^j, 2p^j : j < k\}$ for odd p and $\{0\} \cup \{2^j : j < k\}$ for $p = 2$, so Theorem B' bounds the nullity of every C with $s_W(C) \leq 3$ by the largest element of this set, $2p^{k-1}$ resp. 2^{k-1} , in agreement with [0WE] Theorem 3, whose corank set $\mathcal{N}_p$ is exactly this set without 0.

Theorem T2 (binomials, every $d \geq 2$). $\max\{\dim \ker C : s_W(C) = 2\} = d/p_{\min}(d)$, where $p_{\min}(d)$ is the smallest prime factor of d .

Theorem T3 (trinomials, every $d \geq 3$; conditional on [0WE] Theorem 2). Let $r_3(d)$ be the smallest divisor of d that is ≥ 3 . Then

$$\begin{aligned} \nu_3(d) &:= \max\{\dim \ker C : s_W(C) = 3\} = \max\left(\frac{2d}{r_3(d)}, \frac{d}{2} [d \text{ even}]\right), \\ \max\{\dim \ker C : s_W(C) \leq 3\} &= \max\left(\frac{d}{p_{\min}(d)}, \frac{2d}{r_3(d)}\right). \end{aligned}$$

The second branch of ν_3 is strictly larger than the first exactly when $d \equiv 2 \pmod{4}$ and $3 \nmid d$; it is attained by the anticommuting pair $Z^{d/2}, X^{d/2}$.

Conjecture C (prime powers, all sparsities). For $d = p^k$ and every nonzero C , $\dim \ker C \leq \Omega_k(s_W(C))$.

Conjecture U (every d). For every d and every nonzero C , $\dim \ker C \leq Z_{\leq}(d, s_W(C))$: the monotone collinear maximum bounds all operators. (The exact-sparsity version $\dim \ker C \leq Z(d, s_W(C))$ is false: $d = 10$, $\ell = 3$, Section 5.)

4. Proofs

4.1 Theorem A

Frenkel's lemma ([Fr04]; [3W] Step 6): a nonzero $q = \sum_{i=1}^n a_i x^{t_i} \in \mathbb{k}[x]$ with distinct exponents $t_i \in \{0, \dots, p-1\}$ has $\text{ord}_1 q \leq n-1$. Indeed, with $\theta = x d/dx$, θ maps $(x-1)^a \mathbb{k}[x]$ into $(x-1)^{a-1} \mathbb{k}[x]$, so $\text{ord}_1 q \geq n$ would give $(\theta^s q)(1) = \sum_i a_i t_i^s = 0$ for $0 \leq s < n$, a Vandermonde system in the t_i , which are distinct in $\mathbb{F}_p \subseteq \mathbb{k}$; hence all $a_i = 0$.

We prove $w_p(\text{ord}_1 f) \leq m$ by induction on k , where $\deg f < p^k$ (every polynomial has such a k ; the degree restriction is only used to run the induction). For $k = 0$, f is a nonzero constant, $\text{ord}_1 f = 0$

and $w_p(0) = 1 \leq m$. For $k \geq 1$ write $f = \sum_{j=0}^{p-1} x^j g_j(x^p)$ with $\deg g_j < p^{k-1}$; the exponents $pj' + j$ are distinct, so $m = \sum_j \# \text{terms}(g_j)$. Put $x = 1 + y$. Since $(1 + y)^p = 1 + y^p$,

$$f(1 + y) = \sum_j (1 + y)^j g_j(1 + y^p) = \sum_{n \geq 0} y^{pn} P_n(y), \quad P_n(y) = \sum_j h_{j,n} (1 + y)^j,$$

where $h_{j,n}$ is the coefficient of Y^n in $g_j(1 + Y)$ (the n -th Hasse derivative of g_j at 1) and $\deg P_n \leq p - 1$. The block $y^{pn} P_n$ occupies degrees $[pn, pn + p - 1]$, disjoint for different n . Hence, if $r = \text{ord}_1 f = pr' + r_0$ with $0 \leq r_0 < p$, then $P_n = 0$ for $n < r'$ and $\text{ord}_y P_{r'} = r_0$. Because $\{(1 + y)^j\}_{0 \leq j < p}$ is a basis of the polynomials of degree $< p$, $P_n = 0$ if and only if $h_{j,n} = 0$ for all j . Therefore $\text{ord}_1 g_j \geq r'$ for every j , with equality exactly on $J_0 = \{j : h_{j,r'} \neq 0\} \neq \emptyset$. The polynomial $q(x) = P_{r'}(x - 1) = \sum_{j \in J_0} h_{j,r'} x^j$ is nonzero, has $|J_0|$ terms with distinct exponents in $\{0, \dots, p - 1\}$, and $\text{ord}_1 q = r_0$; Frenkel's lemma gives $|J_0| \geq r_0 + 1$. For $j \in J_0$ the polynomial g_j is nonzero, has degree $< p^{k-1}$ and order r' , so by induction $\# \text{terms}(g_j) \geq w_p(r')$. Altogether

$$m \geq \sum_{j \in J_0} \# \text{terms}(g_j) \geq |J_0| w_p(r') \geq (r_0 + 1) w_p(r') = w_p(r).$$

If $\deg f < p^k$ then $r < p^k$, so $w_p(r) \leq m$ means $r \leq \Omega_k(m)$; conversely $(x - 1)^{\Omega_k(m)}$ has $w_p(\Omega_k(m)) \leq m$ terms (Lucas) and order $\Omega_k(m)$. $\square$

Remark. The function $m \mapsto \max\{\text{ord}_1 f : \text{exactly } m \text{ terms}\}$ is not monotone (for $d = 16$ it takes the value 8 at $m = 2$ and 4 at $m = 3$); the clean statement is the “at most m ” one.

4.2 Alternative proof of the classical scalar law Z

Upper bound. Let $f \in \mathbb{C}[x]$ be nonzero, $\deg f < d$, with at most m terms. Let $K = \mathbb{Q}(\omega)$ (coefficients of f), finitely generated over $\mathbb{Q}(\omega)$, and let $\mathcal{O}$ be an admissible valuation ring of K ; divide f by a coefficient of minimal valuation, so $f \in \mathcal{O}[x]$ and the reduction $\bar{f} \in \mathbb{k}[x]$ is nonzero with at most m terms and degree $< p^k$. The number of zeros of f in μ_d is the nullity of the diagonal matrix $f(Z)$; conjugating by the unnormalised Fourier matrix $F' = (\omega^{jl})$, for which $F'^{-1} Z F' = X$ (Section 4.3), it is also the nullity of the circulant $f(X) \in M_d(\mathcal{O})$. (Reducing $f(Z)$ itself would only give $\bar{f}(1)I$; the circulant form is what makes the reduction informative.) Now $\text{rank}_{\mathbb{C}} f(X) = \text{rank}_K f(X) \geq \text{rank}_{\mathbb{k}} \bar{f}(\bar{X})$, where $\bar{X}$ is the cyclic shift over $\mathbb{k}$. As a $\mathbb{k}[x]$ -module, $\mathbb{k}^d \cong \mathbb{k}[x]/(x^{p^k} - 1) = \mathbb{k}[x]/(x - 1)^{p^k}$, and $\bar{f}(\bar{X})$ is multiplication by $\bar{f} = (x - 1)^r u$ with $u(1) \neq 0$, $r = \text{ord}_1 \bar{f} \leq \deg \bar{f} < p^k$; u is a unit modulo $(x - 1)^{p^k}$ and the kernel of multiplication by $(x - 1)^r$ has dimension r . Hence $\#\{\zeta \in \mu_d : f(\zeta) = 0\} \leq r \leq \Omega_k(m)$ by Theorem A.

Lower bound. Fix $r < p^k$ with digits r_{j-1} and let f_r be as stated. For $n \leq p - 1$ the q -binomial theorem with $q = \omega_p$ gives $\prod_{i=1}^n (u - \omega_p^i) = \sum_{t=0}^n (-1)^t \omega_p^{t(t+1)/2} \binom{n}{t}_{\omega_p} u^{n-t}$, and every Gaussian binomial $\binom{n}{t}_{\omega_p} = \prod_{i=1}^t (1 - \omega_p^{n-t+i}) / (1 - \omega_p^i)$ is nonzero because ω_p is a primitive p -th root of unity and all exponents are in $[1, p - 1]$. So the j -th inner product has exactly $r_{j-1} + 1$ nonzero coefficients, at the exponents ip^{j-1} , $0 \leq i \leq r_{j-1}$. The exponents of the full product are the numbers $\sum_j i_j p^{j-1}$ with $0 \leq i_j \leq r_{j-1}$, which are pairwise distinct base- p expansions below p^k ; no cancellation occurs and f_r has exactly $\prod_j (r_{j-1} + 1) = w_p(r)$ terms and degree r . The factor $z^{p^{j-1}} - \omega_p^i$ ($1 \leq i \leq p - 1$) vanishes at the $z \in \mu_d$ with $z^{p^{j-1}} = \omega_p^i$: the map $z \mapsto z^{p^{j-1}}$ from μ_{p^k} onto $\mu_{p^{k-j+1}}$ has fibres of size p^{j-1} and $\omega_p^i \in \mu_p \subseteq \mu_{p^{k-j+1}}$, so there are exactly p^{j-1} such z , all of exact order p^j . These sets are disjoint over (j, i) , so f_r has at least $\sum_j r_{j-1} p^{j-1} = r$ zeros in μ_d , and exactly r since $\deg f_r = r$. Taking $r = \Omega_k(m)$ gives $Z_{\leq}(d, m) \geq \Omega_k(m)$. $\square$

Note that modulo $1 - \omega$ each factor becomes $(z - 1)^{p^{j-1}}$, so $\bar{f}_r = (z - 1)^r$: the construction lifts the extremal polynomial of Theorem A.

4.3 The Clifford family and the $p = 2$ phases

Let $F' = (\omega^{jl})_{j,l}$, so $F'^{-1} = d^{-1}F'^*$, $F'XF'^{-1} = Z$, $F'ZF'^{-1} = X^{-1}$, hence $F'^{-1}XF' = Z^{-1}$ and $F'^{-1}ZF' = X$ (valid for every d). For odd d let 2^{-1} be the inverse of 2 modulo d and $Q_s = \text{diag}(\omega^{s \cdot 2^{-1} j^2})_j$ for $s \in \mathbb{Z}/d$; for $d = 2^k$ let $\omega_{2d} = e^{2\pi i/2d}$ and $Q_s = \text{diag}(\omega_{2d}^{s j^2})_j$ (well defined on $j \in \mathbb{Z}/d$ because $(j + d)^2 \equiv j^2 \pmod{2d}$). In both cases $Q_s Z Q_s^{-1} = Z$ and $Q_s X Q_s^{-1} = \phi_s X Z^s$ with $\phi_s = \omega^{s \cdot 2^{-1}}$ resp. ω_{2d}^s , and using $(XZ^s)^a = \omega^{sa(a-1)/2} X^a Z^{sa}$ one gets $Q_s W_{(a,b)} Q_s^{-1} = \varepsilon X^a Z^{sa+b}$ with $\varepsilon = \omega^{s \cdot 2^{-1} a^2}$ resp. $\omega_{2d}^{sa^2}$. Put $U_s = F' Q_s$ and $V_s = F'^{-1} Q_s F'$. Then, exactly in $M_d(\mathbb{Q}(\omega))$ resp. $M_d(\mathbb{Q}(\omega_{2d}))$,

$$U_s W_{(a,b)} U_s^{-1} = \varepsilon \omega^{-a(sa+b)} W_{(-(sa+b), a)}, \quad V_s W_{(a,b)} V_s^{-1} = \varepsilon' W_{(a-sb, b)},$$

with $\varepsilon' = \omega^{-s \cdot 2^{-1} b^2}$ resp. $\omega_{2d}^{-sb^2}$ (compute $F' W_{(a,b)} F'^{-1} = \omega^{-ab} W_{(-b, a)}$, apply Q_s , and conjugate back). The first coordinates of the label maps are the linear forms $\lambda_s(a, b) = -(sa + b)$ for U_s ($s \in \mathbb{Z}/d$), $\lambda_\infty(a, b) = a$ for $U_\infty = I$, and $\lambda'_s(a, b) = a - sb$ for V_s . The forms $\lambda_s, \lambda_\infty$ give the $p^k + 1$ primitive directions $(s, 1)$ and $(1, 0)$; the forms λ'_s with $s \in p\mathbb{Z}/d$, $s \neq 0$, give the remaining $p^{k-1} - 1$ directions $(1, t)$ with $p \mid t \neq 0$. Hence the family $\{I\} \cup \{U_s : s \in \mathbb{Z}/d\} \cup \{V_s : s \in p\mathbb{Z}/d \setminus 0\}$ realises each of the $p^{k-1}(p + 1)$ primitive directions exactly once. (This closes the first gap of the review: the family $F' Q_s$ alone, as in [3W], realises only $p^k + 1$ directions when $k \geq 2$; the V_s are needed, and they are what the verification script always used.) All phases lie in μ_d , or in μ_{2d} when $p = 2$.

The phases for $p = 2$. Let $d = 2^k$, $K \ni \omega_{2d}$ and $\mathcal{O}$ admissible. Then $2 \in \mathfrak{m}$, and 2 is a unit multiple of $(1 - \omega_{2d})^{2^k}$ in $\mathbb{Z}[\omega_{2d}]$ (since $1 - \omega_{2d}$ generates the totally ramified prime above 2); as $\mathfrak{m}$ is prime, $1 - \omega_{2d} \in \mathfrak{m}$. So every phase in μ_{2d} reduces to 1, exactly as the d -th roots of unity do. If the coefficient field does not contain ω_{2d} one adjoins it and extends the valuation ring ([St, Tag 00IA]); ranks are unchanged. This closes the second gap of the review.

For each U in the family put $C_U = UCU^{-1} = \sum_g c_g \varepsilon_g W_{M_{Ug}}$: it is an $\mathcal{O}$ -combination of Weyl matrices, hence lies in $M_d(\mathcal{O})$ although U is not unimodular, and $\text{rank}_K C_U = \text{rank}_K C$.

4.4 Theorem B'

Let λ be a primitive direction with $m_\lambda \geq 1$ and U the member of the family whose label map has first coordinate λ . Modulo $\mathfrak{m}$ every Z^b and every phase becomes I , while X becomes the cyclic shift $\bar{X}$ of order p^k over $\mathbb{k}$. Therefore

$$\bar{C}_U = \sum_{t \in \mathbb{Z}/d} \gamma_t^\lambda \bar{X}^t = \bar{g}_\lambda(\bar{X}), \quad \bar{g}_\lambda(x) = \sum_t \gamma_t^\lambda x^t \in \mathbb{k}[x],$$

with exponents taken in $\{0, \dots, d - 1\}$ (legitimate since $\bar{X}^d = I$). The polynomial $\bar{g}_\lambda$ is nonzero, has m_λ terms and degree $< p^k$; as in Section 4.2 the nullity of $\bar{g}_\lambda(\bar{X})$ on $\mathbb{k}^d \cong \mathbb{k}[x]/(x - 1)^{p^k}$ equals $\text{ord}_1 \bar{g}_\lambda$, which is at most $\Omega_k(m_\lambda)$ by Theorem A. Hence

$$\dim \ker_{\mathbb{C}} C = d - \text{rank}_K C_U \leq d - \text{rank}_{\mathbb{k}} \bar{C}_U = \text{ord}_1 \bar{g}_\lambda \leq \Omega_k(m_\lambda) \leq \Omega_k(|S|),$$

the last step because m_λ is at most the number of fibres meeting S and Ω_k is nondecreasing. $\square$

4.5 Elementary proof of the classical affine-plane bound R

Suppose $f \neq 0$ has zero sum on every affine line and $|\text{supp } f| \leq 2p - 1$. A line meeting the support meets it in at least two points (a single nonzero value cannot sum to zero). Call one direction vertical and let $E \subseteq \mathbb{F}_p$ be the set of x whose column $\{x\} \times \mathbb{F}_p$ meets the support; $|E| = p$ would give at least $2p$ points, so $|E| \leq p - 1$, and $E \neq \emptyset$.

Let $R = \mathbb{k}[\mathbb{Z}/p] = \mathbb{k}[t]/(t^p - 1) = \mathbb{k}[u]/(u^p)$ with $u = t - 1$, acting on functions on $\mathbb{F}_p$ by translation, $(tg)(y) = g(y - 1)$, and let $g_x = f(x, \cdot) \in R$. A zero column sum means $g_x \in uR$ (the augmentation ideal). For a slope $\mu \in \mathbb{F}_p$, the sums over the lines $y = \mu x + c$ are the values of $\sum_{x \in E} t^{-\mu x} g_x$ at c , so

$$\sum_{x \in E} t^{\mu x} g_x = 0 \quad \text{in } R \text{ for every } \mu \in \mathbb{F}_p.$$

Let $v = \min_{x \in E} \text{ord}_u(g_x)$; then $1 \leq v \leq p - 1$ and $g_x = u^v g'_x$ with $g'_x(0) \neq 0$ for at least one x . Now $t^{\mu x} = (1 + u)^{\mu x} = \sum_{i < p} \binom{\mu x}{i} u^i$, and for $i < p$ the coefficient $\binom{n}{i} \bmod p$ depends only on $n \bmod p$ (Lucas) and equals $P_i(n)$ for the polynomial $P_i(n) = n(n - 1) \cdots (n - i + 1)/i!$ of degree i over $\mathbb{F}_p$. The coefficient of u^{v+j} ($0 \leq j \leq p - v - 1$) in the identity is $\sum_{x \in E} \sum_{i \leq j} P_i(\mu x) g'_{x, j-i}$, a polynomial in μ of degree $\leq j < p$ that vanishes on all of $\mathbb{F}_p$ and is therefore identically zero; its coefficient of μ^j comes only from $i = j$ and gives $\sum_{x \in E} x^j g'_x(0) = 0$ for $0 \leq j \leq p - v - 1$. If $p - v \geq |E|$ this is an invertible Vandermonde system in the distinct $x \in E$, forcing all $g'_x(0) = 0$, a contradiction; hence $v \geq p - |E| + 1$.

By Frenkel's lemma (the $k = 1$ case of Theorem A: a nonzero element of $u^v R$, viewed as a polynomial of degree $< p$ in t with order $\geq v$ at $t = 1$, has at least $v + 1$ nonzero coefficients), every nonempty column carries at least $v + 1 \geq p - |E| + 2$ support points. In particular $|E| = 1$ is impossible, since a column has only p points (equivalently, $v \geq p$ contradicts $v \leq p - 1$). Thus $2 \leq |E| \leq p - 1$ (so $p \geq 3$; for $p = 2$ we have already reached a contradiction) and

$$|\text{supp } f| \geq |E| (p + 2 - |E|) \geq \min(2p, 3(p - 1)) = 2p,$$

the minimum of the concave function over $[2, p - 1]$ being at an endpoint. This contradicts $|\text{supp } f| \leq 2p - 1$. (The review pointed out that minimising over $[1, p - 1]$ would only give $p + 1$; excluding $|E| = 1$ first, as above, is the missing line.) Sharpness: for parallel lines L_1, L_2 , a line of their class sums to $p \cdot (\pm 1) = 0$ or to 0, and any other line meets each of L_1, L_2 once, giving $1 - 1 = 0$. $\square$

4.6 Lemma N

When $k = 1$, primitive lines are the affine lines of $\mathbb{F}_p^2$, and the conclusion and sharpness follow directly from Lemma R. In the following incidence argument assume $k \geq 2$.

Let $\bar{c} \neq 0$ have zero sum on every primitive line and suppose $|\text{supp } \bar{c}| \leq 2p - 1$. Fix g_0 in the support. Each of the $p^{k-1}(p + 1)$ primitive lines through g_0 contains another support point. A point $h \neq g_0$ with $h - g_0 = p^v h'$, h' primitive, $0 \leq v \leq k - 1$, lies on exactly p^v primitive lines through g_0 : the cyclic subgroups of order p^k containing $p^v h'$ are the $\langle h'' \rangle$ with h'' a lift to $\mathbb{P}^1(\mathbb{Z}/p^k)$ of the class of h' in $\mathbb{P}^1(\mathbb{Z}/p^{k-v})$, and there are $p^{k-1}(p + 1)/(p^{k-v-1}(p + 1)) = p^v$ of them. Let n_1 be the number of support points $h \neq g_0$ with $v = k - 1$, i.e. $h \in g_0 + T_1$, and n_2 the number with $v \leq k - 2$. Counting incidences,

$$n_1 p^{k-1} + n_2 p^{k-2} \geq p^{k-1}(p + 1), \quad n_1 + n_2 \leq 2p - 2,$$

whence $n_1(p - 1) \geq p^2 - p + 2$ and $n_1 \geq p + 1$ (for $p = 2$, $n_1 \geq 4$). So every T_1 -coset that meets the support contains at least $p + 2$ support points; two such cosets would give $2p + 4 > 2p - 1$ points,

so the support lies in a single coset $g_0 + T_1$. A cyclic subgroup H of order p^k meets T_1 in its order- p subgroup $p^{k-1}H$, every order- p subgroup of $T_1 \cong \mathbb{F}_p^2$ arises this way, and a coset of H meets $g_0 + T_1$ either trivially or in a coset of $p^{k-1}H$, i.e. in an affine line of the plane $g_0 + T_1$; conversely every affine line of that plane is such an intersection. Since the support lies in the plane, the primitive-line sums are exactly the affine-line sums of the restriction of $\bar{c}$ to $g_0 + T_1 \cong \mathbb{F}_p^2$, and Lemma R gives $|\text{supp } \bar{c}| \geq 2p$, a contradiction. Sharpness: two parallel lines in $g_0 + T_1$ have zero sum on every affine line of the plane, hence on every primitive line. $\square$

4.7 Theorem B

Let $C = \sum_{g \in S} c_g W_g$ with $|S| = \ell \leq 2p - 1$ and all $c_g \neq 0$. Let $K = \mathbb{Q}(\omega_{2d})(c_g : g \in S)$ (for odd p , $\mathbb{Q}(\omega_{2d}) = \mathbb{Q}(\omega_d)$), a finitely generated field, and let $\mathcal{O}$ be a valuation ring with fraction field K dominating the local subring $\mathbb{Z}[\omega_d]_{(1-\omega_d)}$ ([St, Tag 00IA]). This makes $1 - \omega_d$ lie in its maximal ideal for every prime p . For $p = 2$, Section 4.3 then puts $1 - \omega_{2d}$ in that ideal as well; for odd p , only d -th root phases are used. Scale by a coefficient of minimal valuation. The residue vector $\bar{c}$ is nonzero with support in S , of size at most $2p - 1$, so by Lemma N some primitive line carries a nonzero sum: there is a direction λ with $m_\lambda \geq 1$. Theorem B' gives $\dim \ker C \leq \Omega_k(m_\lambda) \leq \Omega_k(\ell)$, and the closed forms of Section 2 give the explicit bounds. For the sharpness family, $C_\ell = f(Z)$ with $f(z) = \prod_{i=0}^{\ell-2} (z^{p^{k-1}} - \omega_p^i)$; by the q -binomial theorem ($\ell - 1 \leq p - 1$) f has exactly ℓ terms, and it vanishes at the $(\ell - 1)p^{k-1}$ elements of μ_d whose p^{k-1} -th power is one of $1, \omega_p, \dots, \omega_p^{\ell-2}$; so $s_W(C_\ell) = \ell$ and $\dim \ker C_\ell = (\ell - 1)p^{k-1} = \Omega_k(\ell)$. $\square$

4.8 Why $2p - 1$, and what the bound explains

At $\ell = 2p$ the combination $C = \sum_{g \in L_1} W_g - \sum_{g \in L_2} W_g$, for two parallel lines in a T_1 -coset, has integer coefficients whose residue vector lies in the obstruction code of Lemma N: every m_λ vanishes and Theorem B' is silent. The first-order method therefore stops exactly at $2p - 1$; whether the *conclusion* survives is Conjecture C.

For $\ell \leq 3$ the corollary of Section 3 says that the only orders of vanishing available to the reduced circulant are $0, p^j, 2p^j$ (resp. $0, 2^j$): the digit weight explains the shape of the corank set $\mathcal{N}_p$ of [0WE] Theorem 3, though Theorem B' by itself only bounds the nullity by such an order and does not force the nullity to be one. For $p < \ell \leq 2p - 1$ and $k \geq 2$, the operator rank bound p^{k-1} can be strictly stronger than the integral field-free bound $\text{rank } C \geq \lceil d/\ell \rceil$ ([3W] Section 5.2): strictness holds precisely when $\lceil p^k/\ell \rceil < p^{k-1}$. For example, $(d, \ell) = (9, 4)$ and $(25, 6)$ give equality after rounding. Whether $\Omega_k(\ell)$ is attained at *exact* sparsity ℓ in this range is open: the family C_ℓ stops at $\ell = p$, and collinear supports do not attain it (for $d = 27, \ell = 4$ the bound is 18 while the collinear maximum at exact sparsity 4 is 12, Section 5.3). Ω_k is the monotone envelope, and Theorem Z shows that it is attained at exact sparsity $w_p(\Omega_k(\ell))$.

5. Composite dimension

5.1 Binomials

Proof of Theorem T2. Up to a nonzero factor and a phase, $C = c_1 W_g + c_2 W_h = c_1 W_g(I + cW)$ with $W = W_{h-g}$, $c \neq 0$. If $u = h - g$ has order $r \geq 2$ in G , then W^r is a scalar θI and $\text{Tr } W^j = 0$ for $r \nmid j$. The eigenvalues of W are the r roots of θ , with multiplicities m_i satisfying $\sum_i m_i \zeta_r^{ij} = 0$ for $0 < j < r$, so all $m_i = d/r$. The nullity of $I + cW$ is the multiplicity of $-1/c$, i.e. 0 or d/r , and r ranges over the orders

of nonzero elements of $(\mathbb{Z}/d)^2$, all of which divide d ; the maximum $d/p_{\min}$ is attained by $I - Z^{d/p_{\min}}$.
 $\square$ (In polynomial form, $1 + cx^s$ vanishes on a fibre of $x \mapsto x^s$ on μ_d , of size $\gcd(s, d)$; this matches the two-label criterion of [0WE].)

5.2 Trinomials

Proof of Theorem T3, assuming [0WE] Lemma 1 and Theorem 2. After right multiplication by a Weyl adjoint, $D = cI + aU + bV$ with $abc \neq 0$ and $U = W_u, V = W_v$ of distinct nonzero labels. With $UV = \zeta VU$, $q = \text{ord } \zeta$, $H = \langle u, v \rangle$, $h = |\langle qu, qv \rangle|$, [0WE] Lemma 1 gives $|H| = q^2 h$ and a common sector multiplicity $m = d/(qh) \in \mathbb{Z}_{>0}$, and [0WE] Theorem 2 (4b) gives $\dim \ker D \leq m \min(2, h)$. If $h \geq 2$: $\dim \ker D \leq 2d/(qh)$, where qh divides d ; if $q = 1$ then $qh = |H| \geq 3$, and if $q \geq 2$ then $qh \geq 4$; so $qh \geq r_3(d)$ and $\dim \ker D \leq 2d/r_3$. If $h = 1$, then $|H| = q^2$ forces $q \geq 2$ and $\dim \ker D \leq d/q$. If $q = 2$, $q \mid d$ makes d even and gives $\dim \ker D \leq d/2$. If $q \geq 3$, the divisor q satisfies $q \geq r_3(d)$, so $\dim \ker D \leq d/q \leq d/r_3(d) \leq 2d/r_3(d)$. Attainment: (i) let u have order r_3 and choose a phase so that $T = \xi' W_u$ satisfies $T^{r_3} = I$; its eigenvalues are the r_3 -th roots of unity, each of multiplicity d/r_3 ; with $\xi = e^{2\pi i/r_3}$, $(T - I)(T - \xi I) = T^2 - (1 + \xi)T + \xi I$ has three nonzero Weyl coefficients (labels $0, u, 2u$ distinct since $r_3 \geq 3$, and $1 + \xi \neq 0$) and nullity $2d/r_3$ — this is the construction of [0WE] Theorem 3 with p^j replaced by d/r_3 . (ii) If $d \equiv 2 \pmod{4}$, $U = Z^{d/2}$ and $V = X^{d/2}$ are Hermitian involutions with $UV = \omega^{d^2/4} VU = (-1)^{d/2} VU = -VU$, so $(U + V)^2 = 2I$ and $\text{Tr}(U + V) = 0$: $U + V$ has eigenvalues $\pm\sqrt{2}$ with multiplicity $d/2$ each, and $\sqrt{2}I + U + V$ has $s_W = 3$ and nullity $d/2$ (here $q = 2, h = 1, m = d/2$). If $4 \mid d$ then $r_3 \leq 4$ and $2d/r_3 \geq d/2$ already. The “ ≤ 3 ” statement adds Theorem T2 (note $\frac{d}{2}[d \text{ even}] \leq d/p_{\min}$). $\square$

For $d = 10$ and 14 the anticommuting branch wins: $v_3 = 5$, resp. 7 , whereas the collinear maximum at exact sparsity three is $Z(10, 3) = Z(14, 3) = 4$ (exact, Section 5.3). Collinear supports are therefore not extremal at fixed sparsity, but they are extremal in the monotone sense: $\max_{s_W \leq 3} = \max(d/p_{\min}, 2d/r_3)$ is attained on the Z -axis. The [0WE] examples in $d = 12$ and 36 illustrate the sector mechanism and are not maxima: Theorem T3 gives $v_3(12) = 8$ and $v_3(36) = 24$. Numerically (sector formula and SVD, $d \in \{6, 8, 9, 10, 12, 14, 15\}$) the exact-sparsity-three maxima are $4, 4, 6, 5, 8, 7, 10$, all equal to the formula, with the $d = 10, 14$ witnesses anticommuting.

5.3 What fails for composite d , what survives, and the exact tables

If d has two distinct prime factors then $\Phi_d(1) = 1$, $1 - \omega_d$ is a unit and no prime of $\mathbb{Z}[\omega_d]$ has $\omega_d \equiv 1$: the valuation argument has no starting point ([3W] Section 5.4). No additive law in the largest prime factor $p_{\max}$, or largest prime-power factor P , of d can hold: at $\ell = 2$ any bound of the form $(\ell - 1)d/p_{\max}$ or $(\ell - 1)d/P$ is $< d/p_{\min}$, the true maximum of Theorem T2, whenever d is not a prime power ($d = 10$: bound 2, truth 5; $d = 12$: bound 4 or 3, truth 6; at $\ell = 3$, $d = 12$: truth $8 = 2d/3$). The extremal parameters are the smallest divisors $p_{\min}$ and r_3 , in line with Lam–Leung [LL00]: a vanishing sum of m d -th roots of unity exists only if m is a nonnegative combination of the prime factors of d . What survives for all d is the field-free bound $\text{rank } C \cdot s_W(C) \geq d$ and, for collinear supports, Meshulam’s inequality [Me06] on $\mathbb{Z}/d$; the latter is not attained in general: on $\mathbb{Z}/10$ it allows 6 zeros at $m = 3$ and 7 at $m = 4$, while the exact values are 4 and 6.

Exact tables. $Z(d, m)$ was computed by enumerating, for each support S with $|S| = m$ up to the affine group of $\mathbb{Z}/d$ (units and translations), the flats cut out by at most $m - 1$ character rows over $\mathbb{Q}(\omega_d)$, with arithmetic in $\mathbb{Q}(\omega_d)$ realised through the regular representation and kernels over $\mathbb{Q}$; a flat carries a full-support vector if and only if no coordinate vanishes identically on it, and its closure is then a zero set. (Correctness: a full-support f with zero set T spans a subspace of dimension $\leq m - 1$ of row

conditions, so some $R \subseteq T$ with $|R| \leq m - 1$ cuts the same flat, and every full-support vector of that flat vanishes on the closure.) The reviewer recomputed the values by a different exact method (rank criterion over pairs (S, T) , refuting $v + 1$ zeros and attaining v , with python-flint). Only the entries certified in this way are called exact; the remaining entries come from the earlier finite-field model ($\mathbb{F}_q$ with two independent primes $q \equiv 1 \pmod d$, which can only under-estimate ranks and is therefore not a certificate in either direction at fixed m , although two agreeing primes make an error very unlikely). The working report had labelled all of them exact; that label is withdrawn here.

d	exact over $\mathbb{Q}(\omega_d)$ ($m = 1, 2, \dots$)	$\mathbb{F}_q$ only (not certified)
6	0, 3, 4, 4, 4, 5 ($m \leq 6$)	—
10	0, 5, 4, 6, 8, 7, 6, 8, 8, 9 ($m \leq 10$)	—
12	0, 6, 8, 9, 8, 10, 8 ($m \leq 7$)	$m = 8$: 10
14	0, 7, 4, 8, 8 ($m \leq 5$)	—
15	0, 5, 10, 9, 12 ($m \leq 5$)	$m = 6, 7$: 11, 9
18	0, 9, 12, 12 ($m \leq 4$)	—

For $d = 15$ the values $m \leq 3$ follow from Theorems T2, T3 (collinear attainment), $m = 4$ is the reviewer's exact spot check, and $m = 5$ is an exact run of the author's flats script made for this manuscript (34 support orbits, 4455 s; repro/out/zerosQ_cyclic_15_m5.out). Entries for $d = 20, 21, 30$ mentioned in the working report have no log and are dropped. Constructions matching the exact entries: $m = 2$: a fibre of $x^{d/p_{\min}}$; $m = 3$: $2d/r_3$ ($d = 6, 12, 15, 18$) or, for $d = 10, 14$, the value 4; $m = 4$: $3d/r_4$ with r_4 the smallest divisor ≥ 4 ($d = 12$: 9; $d = 18$: 12), or the product $(1 + ax^{d/r})(1 + bx^{d/s})$ with $\gcd(r, s) = 1$, which has $d(r + s - 1)/(rs)$ zeros ($d = 6$: 4; $d = 10$: 6; $d = 14$: 8); $m = 5$, $d = 10$: $(m - 1)d/5 = 8$. No closed formula for general m is claimed.

For prime powers the running maxima are Theorem Z; the exact-sparsity values are not monotone. For $d = 16$ the values $Z(16, 3) = 8$ and $Z(16, 4) = 12$ are certified exactly (reviewer), and $Z(16, m) = \Omega_4(m)$ for $m \leq 2$ by Theorem Z; the $\mathbb{F}_q$ model gives 8, 12, 12 for $m = 5, 6, 7$. For $d = 27$, $Z(27, m) = \Omega_3(m)$ for $m \leq 3$ by Theorem Z, and $Z(27, 4) = 12$ is certified exactly by the reviewer's flats method (Section 9) — this is the value used in Section 4.8.

6. Conjecture C and the evidence

6.1 Where the conjecture is open

For $d = p^k$ and a sparsity ℓ , Conjecture C is implied by Theorem B when $\ell \leq 2p - 1$ and by the field-free bound when $\Omega_k(\ell) \geq d - \lfloor d/\ell \rfloor$. The remaining sparsities, computed by two independent scripts (author and reviewer, identical output), are:

d	Theorem B covers $\ell \leq$	open ranges of ℓ (rank that must be proved)
4, 8	3	none: Conjecture C is a theorem
9	5	none: Conjecture C is a theorem
16	3	[6, 7] (rank ≥ 4 ; field-free gives 3)
25	9	[10, 14] (≥ 4); [15, 19] (≥ 3)

d	Theorem B covers $\ell \leq$	open ranges of ℓ (rank that must be proved)
27	5	[6, 8] (≥ 6); [14, 17] (≥ 3)
32	3	[5, 7] (≥ 8); [11, 15] (≥ 4)
49	13	[14, 20] (≥ 6); [21, 27] (≥ 5); [28, 34] (≥ 4); [35, 41] (≥ 3)

The first open case overall is $d = 16, \ell = 6$: is every $C \in M_{16}(\mathbb{C})$ with six or seven nonzero Weyl coefficients of rank at least 4?

6.2 Torsion cosets

Let $2j \leq k$ and let the support lie in a coset of T_j . After a translation the support lies in $T_j = p^{k-j}G$, where $\langle g, h \rangle \in p^{2(k-j)}\mathbb{Z} = 0$ in $\mathbb{Z}/d$, so $g \mapsto W_g$ is an honest representation of $T_j \cong (\mathbb{Z}/p^j)^2$; by trace orthogonality each character χ occurs with multiplicity p^{k-2j} , and $C = \sum_{\chi} \hat{c}(\chi) P_{\chi}$ with $\hat{c}(\chi) = \sum_g c_g \chi(g)$. Hence $\dim \ker C = p^{k-2j} \cdot \#\{\chi : \hat{c}(\chi) = 0\}$, a commutative uncertainty problem on $(\mathbb{Z}/p^j)^2$.

p-torsion ($j = 1, k \geq 2$). Meshulam's inequality on $\mathbb{F}_p^2$ with the consecutive divisors $1, p, p^2$ gives, for $\ell \leq p$, at most $p(\ell - 1)$ zero characters, and for $p \leq \ell \leq p^2$ at most $p^2 - p - 1 + \lfloor \ell/p \rfloor$; multiplied by p^{k-2} these are exactly $(\ell - 1)p^{k-1}$ and $(p - 1)p^{k-1} + (\lfloor \ell/p \rfloor - 1)p^{k-2}$, i.e. $\Omega_k(\ell)$ (with $\Omega_k(p^2) = p^k - p^{k-2}$ at $\ell = p^2$). So Conjecture C holds for every support in a coset of the p -torsion, for every $k \geq 2$. (Re-derived by the reviewer; the comparison script reports no case with $j = 1$ in which the Meshulam bound exceeds $\Omega_k(\ell)$, over all $p \leq 7, p^k \leq 2401$ and $\ell \leq p^2$, i.e. 316 cases.)

4-torsion in $d = 16$ and 32 ($p = 2, j = 2$). Here Meshulam's bound on $(\mathbb{Z}/4)^2$ is slack (it allows $10 > 8$ zeros at $\ell = 3$ and $13 > 12$ at $\ell = 6, 7$), so exact values are needed. The exact maxima of zero characters on $(\mathbb{Z}/4)^2$ for full-support functions with exactly m points are 0, 8, 8, 12, 8, 12, 10 for $m = 1, \dots, 7$ (author, exhaustive over the orbits of the affine group of order 1536; reviewer, independent method, $m \leq 6$). Fourier duality closes the rest: $|\text{supp } f| \in \{2, 3\}$ forces $|\text{supp } \hat{f}| \geq 8$, hence (applying this to $\hat{f}$) $|\text{supp } \hat{f}| \in \{2, 3\}$ forces $|\text{supp } f| \geq 8$, and $|\text{supp } \hat{f}| = 1$ forces $|\text{supp } f| = 16$. Therefore $|\text{supp } f| \leq 7$ gives $|\text{supp } \hat{f}| \geq 4$, at most 12 zeros, and $8 \leq |\text{supp } f| \leq 15$ gives at most 14 zeros. Since $\Omega_4(m) = 12$ for $4 \leq m \leq 7$ and 14 for $8 \leq m \leq 15$, Conjecture C holds for every support in a coset of T_2 in $d = 16$ (multiplicity 1), and, with multiplicity $2^{5-4} = 2$, in $d = 32$: the bounds 16, 24, 28, 30 for $m \in \{2, 3\}, [4, 7], [8, 15], \{16\}$ equal Ω_5 there.

6.3 Numerical searches and a false alarm

Alternating projections between the span of the support Weyls and the variety of rank $\leq d - \kappa$ matrices were run in the open ranges: $d = 16, \ell \in \{5, 6, 7\}$, target nullity 13 (300 random and 100 “blocked” supports containing a T_1 -coset, 6 restarts, 300 iterations each); $d = 27, \ell \in \{6, 7, 8\}$, target 22; $d = 25, \ell \in \{10, 12, 14\}$ (target 22) and 16 (target 23). No full-support witness was found (best relative residuals $\geq 0.46, 0.35, 0.18$). Control searches against the proved statements ($d = 9, \ell = 4, 5$; $d = 8, \ell = 3$; $d = 16, \ell = 3$; $d = 25, \ell = 8, 9$; $d = 27, \ell = 4, 5$) likewise found nothing, as Theorem B requires. These are finite searches, not evidence of the same kind as Sections 6.1–6.2.

Three instances in $d = 16$ had been flagged numerically as having nullity 1 where Theorem B' predicts 0 (a single coefficient survives modulo 2). Exact computation of the rank of the 128×128 integer matrix of the regular representation gives 128 in all three cases, so $\det C \neq 0$; the matrices are merely

ill-conditioned ($\sigma_{\min}/\sigma_{\max}$ between 10^{-16} and 10^{-12} , the sector determinant $c^{16} - (a^{16}s + b^{16}t)$ nearly cancelling). The reviewer confirmed the first instance by hand: its relative labels have commutator of order 16, one sector, and determinant $1 - 8^{16}(s + t)$ with s, t roots of unity, which cannot vanish. No violation of Theorem B' exists.

7. Verification

The following is the author-supplied historical coverage, with each author's or reviewer's scope stated individually. These labels describe supplied scripts and logs; they do not authenticate historical model identity or contextual independence. Some results were checked by separately written programs, some are author-only, and some slow runs were not repeated (Section 9 and the reproduction README). Exact certifications and floating-point diagnostics are distinct.

1. *Theorem A*. Exhaustive over exponent sets (up to translation, resp. up to the affine group of $\mathbb{Z}/p^k$), with the maximal order over full-support polynomials decided by a rank criterion on the Lucas/Hasse matrix over $\mathbb{F}_p$ (the statement over the algebraic closure): $d = 16$ all m ; $d = 25, 27$ for $m \leq 7$ (author); $d = 4, 8, 9, 16$ all m and $d = 25, 27$ for $m \leq 7$ (reviewer, both the exact-order form $w_p(\text{ord}) \leq \# \text{terms}$ and the monotone form). The non-monotone exact maxima are reproduced ($d = 16$: exactly 3 terms give at most order 4, exactly 2 give 8).
2. *Theorem Z*. Exact certification in $\mathbb{Z}[\omega_d]$ (arithmetic modulo Φ_d) that f_r has exactly $w_p(r)$ terms and at least r zeros in μ_d , for every $r < d$ and $d \in \{4, 8, 9, 16, 25, 27, 32, 49, 81\}$ (author; 1–95 s per d); exactly r zeros for $d \in \{4, 8, 9, 16, 25, 27\}$ (reviewer, own arithmetic). Also the sharpness family C_p .
3. *Clifford identities*. Author: all conjugation identities for U_s and V_s checked numerically (complex doubles, discrete target, tolerance 10^{-9}) for $d = 16, 25, 27$ (4352, 15625, 19683 identities): no failure. Reviewer: exact verification in $\mathbb{Z}[\omega_{2d}]$ ($p = 2$) or $\mathbb{Z}[\omega_d]$ for $d = 4, 8, 9$ (96, 768, 972 identities), including that the phases lie in μ_{2d} resp. μ_d and that the family realises exactly 6, 12, 12 directions.
4. *Lemmas R and N*. Reviewer: minimum weight of the Radon code on $\mathbb{F}_p^2$ for $p = 2, 3, 5$ and of the obstruction code $N(p, k)$ for $d = 4, 8, 9, 16$ equals $2p$ (meet-in-the-middle over column combinations, exact over $\mathbb{F}_p$, field-extension invariant). Author: $\dim N$ over $\mathbb{F}_p$ is 7, 34, 45, 160, 400, 489 for $d = 4, 8, 9, 16, 25, 27$, the two-line word lies in N , the full weight distribution for $d = 4$ (127 words, minimum 4), and an independent minimum-distance check for $d = 8, 16, 9$.
5. *Theorem B' on instances*. 300 ($d = 16$) and 200 ($d = 25, 27$) integer-coefficient instances, comparing the numerical nullity with $\text{ord}_1 \bar{g}_\lambda$ in every primitive direction; the three flags were resolved exactly (item 6.3).
6. *Theorems T2, T3*. Sector formula plus SVD for $d \in \{6, 8, 9, 10, 12, 14, 15\}$ (author); exhaustive binomial maxima for $2 \leq d \leq 15$, 60 random trinomials against the sector formula, and the attainment examples (reviewer).
7. *Tables and gap ranges*. Section 5.3 and 6.1: author and reviewer values agree wherever both exist; the gap table and the closed forms of Ω_k were recomputed independently for all listed d .

Fresh local checks, 14 September 2026. The new original-PDF reviews checked the full arguments A, Z, B', R and N, executed exact root constructions and Clifford checks for small prime powers, and rechecked selected incidence-code minima and exact tables. The scalar Hamming-number equivalence was checked against the primary source. They did not replay every large table, all high-dimension obstruction codes or all low-rank searches. This candidate preserves all supplied Python and data bytes, and adds an exact integer check of the new scalar-formula translation and the rounded-bound

case split. Original and new execution records are kept separately. None of the finite tests establishes the open all-sparsity conjectures.

8. Open questions

1. **Conjecture C** at $d = 16$, $\ell \in \{6, 7\}$. A proof would need either a second-order (Newton-polygon) refinement of Theorem B' for residue vectors in the obstruction code, or an exact exhaustive computation over the $AGL(2, \mathbb{Z}/16)$ -orbits of 6- and 7-point supports, which is within reach of the flats method.
2. **Exact sparsity.** Is $\Omega_k(\ell)$ attained at exact sparsity ℓ for $p < \ell \leq 2p - 1$? The collinear answer is no ($d = 27$, $\ell = 4$: $12 < 18$); the non-collinear answer is unknown, and by [OWE] Theorem 3 the coranks at $\ell = 3$ are constrained to $p^j, 2p^j$.
3. **Composite** d , $\ell \geq 4$. Conjecture U and a closed form for $Z(d, m)$; the product constructions of Section 5.3 and the values $3d/r_4$ suggest that the smallest divisors continue to govern, but $Z(10, 5) = 8 = 4d/5$ and $Z(15, 5) = 12 = 4d/5$ are already of the form $(m - 1)d/r$ with $r = 5$ the *largest* divisor below $m + 1$.
4. **The obstruction code.** Determine the minimum-weight words and the weight distribution of $N(p, k)$ beyond the two-line words; a characterisation of low-weight words would tell exactly where Theorem B' can and cannot be applied.
5. **Other groups.** The proof uses only that $1 - \omega$ is a prime with residue characteristic p and that the Clifford family covers all primitive directions; the same argument should apply to Weyl systems over $(\mathbb{Z}/p^k)^{2n}$ and, with Meshulam's inequality replacing Theorem A, to groups of order pq in the spirit of [Lo25].

9. Reproducibility

The directory `repro/` next to this manuscript contains: `author/` (the thirteen scripts of the working report, unchanged, with their logs in `author/logs/`); `reviewer/` (the reviewer's nine scripts and two logs, unchanged); `run_fast.py`, which reruns the fast subset and writes `out/<name>.out` and `out/runtimes.txt`; and `README.md` with provenance and the list of slow runs that were not repeated (the exact tables for $d = 10, 12, 14, 18$ and the plane $(\mathbb{Z}/4)^2$ beyond $m = 4$, the $d = 25, 27$ obstruction codes, the numerical searches). Python 3.12, numpy 2.5.1, sympy 1.14, python-flint 0.9.0 (reviewer's exact scripts only). The rerun of 13 September 2026 is summarised in `README.md`; the fast jobs reproduced their logged outputs, including the known floating-point threshold mismatch at $d = 14$ in `gap_table.py`; exit status zero does not mean every numerical comparison agreed.

AI-assistance statement

The author supplied the original manuscript and code as work developed with Claude Fable 5.1, with author, adversarial-review and writing roles. Those historical model identities, configurations, roles and session independence have not been authenticated here. Native Codex tasks configured as GPT-6 Astra High and GPT-5.6 Sol Medium made fresh local assessments of the original exact PDF on 14 September 2026; both identified the scalar prior-art equivalence. The coordinator prepared this revised candidate with explicit scalar attribution, corrected valuation and case-split arguments, and coverage qualifications. The original assessments remain attached to the original hash; they do not assess or accept this new PDF. Theorem A is classical [MCJ73], Theorem Z is [DVB06, DVB08] in equivalent notation, and T3

remains conditional on [OWE]. No independent human peer review is claimed. Scientific responsibility rests with the author.

References

- [3W] L. Eriksson, *Rank plus Weyl sparsity is at least $p + 1$: an operator form of the Chebotarev–Tao uncertainty principle in prime dimension*, AI Research Record, ARR-2026-3VWVP20DY09HFASA (2026). Cited: Sections 2–5 (setting, Theorem 1 and 1', Steps 1–6, Sections 5.2–5.4), Open questions 1 and 3, Corollary 2.
- [OWE] L. Eriksson, *Three-term Weyl operators: central multiplicities, rank laws, and exact list decoding*, AI Research Record, ARR-2026-0WESHW4YMM9FG8EK (2026). Cited: Lemma 1, Theorem 2 (eq. (4b)), Corollary 2.1, Theorem 3, Proposition 4 and eqs. (6a), (6c), eq. (8).
- [3H0] L. Eriksson, *Strictly scalable exterior decoders for quantum lists: exact full-spark widths and fixed-probe Weyl Bayes curves*, AI Research Record, ARR-2026-3H0ZKWJMH18MH9FX (2026). Cited: Theorem 6.1, Remark 6.4, Corollary 6.5.
- [MCJ73] J. L. Massey, D. J. Costello, J. Justesen, Polynomial weights and code constructions, *IEEE Trans. Inform. Theory* IT-19 (1973) 101–110. Cited: Theorem 1.1, Section III-B/Theorem 6.1, Lemma 1 (weight-retaining property, $W[(x - c)^i] = \prod (i_j + 1)$).
- [Me06] R. Meshulam, An uncertainty inequality for finite abelian groups, *European J. Combin.* 27 (2006) 63–67.
- [Ta05] T. Tao, An uncertainty principle for cyclic groups of prime order, *Math. Res. Lett.* 12 (2005) 121–127.
- [Fr04] P. E. Frenkel, Simple proof of Chebotarev’s theorem on roots of unity, arXiv:math/0312398 (2004).
- [DVB08] S. Delvaux, M. Van Barel, Rank-deficient submatrices of Fourier matrices, *Linear Algebra Appl.* 429 (2008). DOI: [10.1016/j.laa.2008.04.043](https://doi.org/10.1016/j.laa.2008.04.043).
- [DVB06] S. Delvaux, M. Van Barel, *Rank-deficient submatrices of Fourier matrices*, Report TW470 (September 2006), [author-hosted text](#). Definition 7 and Theorem 10, equation (23), use a monotone support budget.
- [KMM08] J. D. Key, T. P. McDonough, V. C. Mavron, *An upper bound for the minimum weight of the dual codes of desarguesian planes*, [author-hosted postprint](#), 17 January 2008, Section 2, Result 1 (the affine-plane prime-order bound).
- [LL00] T. Y. Lam, K. H. Leung, On vanishing sums of roots of unity, *J. Algebra* 224 (2000) 91–109.
- [BL18] A. Bíró, V. F. Lev, Uncertainty in finite planes, arXiv:1808.07424 (2018).
- [Lo25] M. Loukaki, Chebotarev’s theorem for cyclic groups of order pq and an uncertainty principle, *Bull. Lond. Math. Soc.* (2025), arXiv:2412.08600.
- [Lu78] E. Lucas, Théorie des fonctions numériques simplement périodiques, *Amer. J. Math.* 1 (1878) 184–240, 289–321.
- [St] The Stacks Project Authors, *The Stacks Project*, Tag 00IA (extension of valuation rings).